

8 The Need for Multipolar Artificial Intelligence Governance

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8.1 Introduction

Advances in artificial intelligence (AI), and the amount of financial resources invested in newer models, are increasing at a breathtaking pace. Even considering that many of the rumoured capabilities and promises of new AI models are only hype, there is no doubt that AI will impact lives and the global economy for decades to come.¹

These breakthroughs—and the potential use of this rapidly changing technology for both positive and negative purposes—mean that the governance of AI (i.e. how these technologies should be developed, used, and regulated, and how consumers and workers should become more informed about them) has become a central topic of discussion amongst social scientists and AI experts.² AI governance is partly about technical issues, such as assigning liability for misuse or malfunction, addressing reliability issues (e.g. hallucinations), and tracking progress and potential damages from the technology especially when AI is used in consumer-facing ways.³ Yet often ignored in the technocratic debates is that governance is also about politics. Who has a voice in the development and adoption of new technology? Who will control these new AI tools?

As many authoritarian governments and especially China's Communist Party understand, whoever controls information will gain disproportionate social and political power.⁴ It is also true that whoever controls the use of new technologies in production will gain a significant amount of power. Tech leaders—hoping to shape both information and technology deployment—promise that AI will remake the division of labour around the world. Yet who will benefit from this upheaval?

Currently, there is a serious imbalance of power over the design and diffusion of AI. This imbalance is exacerbating the current socially detrimental trajectory, philosophy, and usage of AI technologies. It is also undermining any kind of organic guardrails that could emerge from greater competition or countervailing forces of worker voices and civil society. Despite growing concerns about what AI will mean for everyday lives, this nascent technology is still controlled only by a few companies and executives in Silicon Valley and a similarly small number of companies and government leaders in China, with a secondary role for US regulators and the European Union. Better rules must be constructed for AI

governance, and there have been some promising ideas. Yet ultimately, the politics of governance will matter. Political outcomes depend, at least in part, on whose voices will be heard and whose will be stifled.

On the current trajectory of AI, two groups will bear the brunt of the consequences, especially if events go poorly—(i) workers in the industrialised world and (ii) citizens in the developing world. Neither of those two groups currently has any voice or power over the future of AI, however.

This chapter examines problems with the current direction of AI and possible solutions. It then posits why multi-polarity is important and discusses how the current environment—in which the poles are the tech companies, Government of China, US regulators, and EU policymakers—is not sufficient to make AI work for everyone. The importance of worker voices is stressed, and that the developing world is completely absent from the debate is emphasised. These themes are united in the conclusion, which also offers concrete policy ideas.

8.2 Why AI Governance Is Central: Promise and Peril

AI is an information technology; it can collect, process, and deploy information. As an information technology, AI offers new possibilities. Today, information is abundant (i.e. most human knowledge—and a lot of misinformation—is available on the internet), but useful information (i.e. that needed to reliably and immediately perform complex decision-making and problem-solving tasks) is scarce. AI tools, especially recent generative AI technologies, could be deployed to provide real-time, reliable, and context-dependent information to human decision makers.

The potential of AI is not just for office workers, academics, and journalists, as modern craft workers (e.g. electricians, plumbers, carpenters, and maintenance workers) also routinely navigate a series of problem-solving tasks in their jobs. Yet there is a lack of accessible information to handle the complex problems that such workers repeatedly encounter. As an example, electricians may be trying to correct issues with new electrical equipment or addressing unanticipated challenges from a quickly evolving electricity grid. Access to better information can enable these workers to attain and to apply greater expertise in a range of activities and to take on new, more sophisticated tasks. Past literature has explored and emphasised this potential avenue for pro-worker AI.⁵

This promise is particularly attractive because it could also help reverse the concerning trends of occupational and wage polarisation that have pervaded the United States and other industrialised nations since the late 1970s. Middle-income jobs have become more scarce; thus, displaced workers have been pushed disproportionately into low-wage jobs.⁶ One major driver of this labour market bifurcation has been the preponderance of industrial automation (i.e. the substitution of capital for labour, reducing the need for workers in the production process by deploying industrial robots).⁷ This automation-substitution tends to increase inequality and lead to only slow—or sometimes even negative—wage growth. This downward pressure could be relieved if new, productive tasks are created for workers to help reduce inequality and boost wages.⁸ It is possible—although

not inevitable—that deploying new AI tools could increase the demand for middle-skilled labour, enabling such workers to perform more expert work and allowing them to attain better compensation in return for their enhanced productivity.

AI, like other information tools, can also be used to further accelerate automation (i.e. deploying information for finding ways of mimicking human performance in various tasks). If so, it is likely to deliver the same type of inequality-boosting growth experienced in the United States over the last 4 decades. AI has the capability to automate white-collar tasks, and, in the future, it could be integrated with other technologies, including robotics, to deepen the automation of blue-collar tasks.⁹ Moreover, AI's information can also be manipulated and distorted—as experienced with social media over the last 2 decades. It could also increase the effectiveness of digital ads and filter bubbles at a steep social cost.

The information provision capabilities of AI can also be used in new, pro-democratic ways. AI that provides more reliable information; increases transparency about government activities; creates new pathways for citizen participation in public decision-making; and builds new public spheres for debate, deliberation, and information exchange can be a boon to democratic citizenship.¹⁰ Taiwan, which has rolled out AI tools to increase governmental transparency, bolstering citizen voices and participation by democratising access to and control of information, illustrates one pathway for this achievement.¹¹

AI embodies a way to rebuild shared prosperity and democratic citizenship along one path, or further reinforce inequality and deepen the damage to democracy and well-being along another.¹² Unfortunately, there is ample evidence that the latter path is much more likely. First, the business models of tech companies shepherding AI advances are very similar to those that led to prioritising automation and information manipulation in the past. It is notable that the two areas that have received the most attention from the industry are (i) automation of additional white-collar tasks using generative AI and (ii) internet search and related activities, where profits come from digital ads.¹³

Second, advances in AI have intensified notions of artificial general intelligence, and these ideas have played an important role in shaping the approach to generative AI that has emerged (e.g. based on very large foundation models that mimic human conversation).¹⁴ If it is perceived that human-level cognition and capabilities are soon coming, then it becomes natural—or even inevitable—to automate more tasks, so that the 'more efficient' AI models perform them instead of inconsistent humans. Moreover, if it is believed that AI will soon be better than much of humankind, it becomes much less problematic to accept algorithmic dominance over humans so that algorithms guide human decisions and, when necessary, manipulate those decisions (and extract abundant data for use without the need for permission or recompense).

Third, early evidence suggests that the AI playing field may be much more oligopolistic (i.e. dominated by two or three leading companies with similar visions and business models) than previous digital technologies.¹⁵ Such an oligopolistic structure entails higher prices and a slower spread of beneficial AI applications.

More ominously, it could lock in a future path of AI development that is more pernicious—more focussed on only automation and more manipulative—while simultaneously making it harder for alternative business models that could usher in greater human complementarity and more pro-democratic uses of AI to become competitive.

If this perilous direction is indeed on the horizon, what are the options? The simple answer is to redirect the trajectory of AI. This will require not only technocratic regulation but also a broader mobilisation of society and changes in norms. This motivates the importance of multi-polarity.

8.3 The Centrality of Multi-Polarity

A multipolar governance structure for AI is needed. There is an imperative to redirect AI in a more socially beneficial direction, but such a challenge cannot be accomplished by just attempting unsystematic, ad hoc solutions to social symptoms as they arise. It requires new norms and a different balance of power that guides AI priorities towards what is broadly beneficial for society, not just a few select technologists.

Multi-polarity does not mean sidelining AI expertise. Of course, tech executives' advice and visions need to be considered, as they know much about the technology and make significant decisions about where it will go. Equally, any regulation would fail if it were not partly designed and enforced by mission-driven (i.e. democracy-driven) governmental experts.

More poles and more voices are needed, in part, because—despite apparent differences—there is also congruence between the approaches of the two AI superpowers, the United States and China. Implicit acceptance of algorithmic dominance over humans is common both in the United States and China. Leading players in both countries are committed to the advancement of digital technologies and the use of AI as a tool for collecting information from humans.¹⁶ Moreover, companies and policymakers in both countries view algorithmic automation as the engine of future economic growth. In the United States, algorithmic dominance takes the form of large platforms collecting data and manipulating people, while in China, tech companies are empowered if they follow the government's agenda.¹⁷ This difference notwithstanding, neither the United States nor China is likely to strengthen the privacy and protection of citizens or consumers. Nor are they likely, without further pressure, to pour resources into redirecting the trajectory of AI research in a more human-complementary direction.

To counter algorithmic dominance over humans, small, regulatory fixes will not work. Even if the right types of regulations were adopted, they would not be wholly effective unless norms in the tech industry also evolve. Yet norm changes require politics.¹⁸ In addition, giving a voice to so-far ignored stakeholders would be an important step towards favouring the human perspective.

The European Union has played a leading role in regulating AI to date. It has been at the forefront of privacy protection against tech companies and anti-trust activity.¹⁹ Top-down regulation, without changing the norms and power imbalance

of the tech industry, can only go so far, however. Moreover, Europe does not house any of the leading AI companies. Of the six digital services gatekeepers specified by the 2022 EU Digital Markets Act, five are US companies (i.e. Alphabet, Amazon, Apple, Meta, and Microsoft) and the sixth is Chinese (i.e. ByteDance). The European Union's comparative advantage in regulating originates exactly from this absence of big powerful players, who would otherwise lobby against regulations. This also limits what the European Union can achieve since it cannot effectively embed those policies and regulations into the normative fabric of a wide-ranging ecosystem to encourage a different AI research direction.

8.4 Worker Voices

Worker voices are important for two reasons. First, organised labour can be a valuable counterweight to the emphasis on automation and surveillance. Its role can be particularly important when managers have incentives to use new technologies for automation—either because they believe it is the most effective business strategy or they are motivated by desires to undercut the power of labour. Labour unions can resist excessive automation and encourage firms to introduce new tasks and training opportunities for their workers.

There have been instances in the past where organised labour in the United States has succeeded in influencing the rollout of new technology and creation of new tasks for workers.²⁰ Notable examples include the integration of new numerically controlled tools in the automotive industry in the 1950s and 1960s, and more recently, the 2023 Writers Guild of America negotiations over the use of generative AI tools for scriptwriting.²¹ The approach of negotiating over the terms of technology instead of only downstream issues like wages has been more common in Scandinavia and Germany. Nordic unions have often worked with companies on worker compensation and on how work itself is organised, while work councils and labour representatives on corporate boards have often represented workers' viewpoints in the process of technology adoption.²²

Moreover, the importance of labour voices is even more central when it comes to AI, because, as previously articulated, AI is an information tool with the promise of improving how workers engage in problem-solving tasks. Trying to improve how workers use information and engage in problem-solving without actually consulting workers would be like trying to solve the problem of child malnutrition in a village in Malawi without ever finding out anything about Malawi or the village.²³ The real promise of worker voices is that they can provide context, as well as insight, about what types of information would be most useful to workers, as well as the training that workers would need to incorporate emergent AI tech capabilities.

Part of the reason why the current direction of AI is prioritising automation and not increasing worker productivity is that there are currently no worker voices in AI development and adoption. Robust worker voices used expertly and correctly—not as a barrier to technology adoption but to steer the development of workplace technology in a more pro-worker direction—could determine which alternative becomes the future of AI and jobs.

Robust worker voices necessitate more powerful labour organisations as well. It is not clear whether existing labour unions can play this role or if new organisational forms must develop. More bottom-up participation from workers is needed. Furthermore, the labour movement itself may have to change, becoming both more invested in AI expertise and more flexible about how it approaches technology. In short, the labour movement must find the right posture that is both accommodating to new technologies and forceful in articulating ways of using these technologies in more pro-worker ways.

8.5 Developing Country Voices

AI is a global technology. If its adoption is significant in the industrialised world, it will almost surely spread to the developing world. A glimpse of this is already apparent in AI technologies used for surveillance. Existing evidence indicates that as the Government of China intensifies data collection and increases its demand for technologies that process these data and use them for surveillance and monitoring, this is having a first-order impact on the direction of AI research amongst Chinese startups. This effect is not confined to China; these surveillance technologies are spreading to other countries, especially authoritarian ones that have a greater demand for citizen-monitoring capabilities.²⁴ The Chinese telecom giant Huawei, for example, is reported to have exported surveillance technologies to at least 50 other countries (Feldstein, 2019). Many Western countries are also using sophisticated AI tools for surveillance of their citizens.²⁵

The case of surveillance technologies illustrates not just the inevitable spread of new AI tools around the world but also the more extreme dangers that some of these tools pose in countries with weaker institutions. Courts, civil society pressure, and the democratic process generate—albeit imperfect—barriers against limitless use of these tools by governments. Such checks and balances are much less likely to be forthcoming in authoritarian countries around the world, exacerbating concerns of abuse.²⁶ The same concerns apply to data collection and how AI-based technologies could be abused in workplaces (e.g. to reduce breaks or to increase the speed of work).²⁷

AI adoption in the industrialised world would impact developing countries even if firms in those places did not invest in this technology because AI uptake in Western countries will inevitably remake the global division of labour. This raises another set of major concerns. If AI goes in the direction of automation, it may encroach into the set of tasks that could have been profitably performed in middle-income economies, especially as the education level of the workforce in these countries continues to improve. This could significantly reduce the remaining opportunities for rapid economic growth in those places, in a similar pattern to the premature de-industrialisation driven by globalisation and industrial automation that has held many developing countries in lower income levels (Rodrik, 2015).

The most prominent examples of fast growth amongst emerging economies (e.g., China, Malaysia, South Korea, and Viet Nam) over the last 6 decades occurred when these countries found a niche in the global division of labour by exporting products or tasks where their abundant labour supply created a comparative advantage,

then upgrading their skills and competencies to move to the next rung of the value chain. If the goods in which these countries initially specialised (e.g. textiles, apparel, or toys) were automated cheaply at the time, then these growth opportunities would not have existed. Thus, AI focussing predominantly on automation may have similar effects for the next group of potential first-growers.

Currently, most of those who will be impacted by AI (around 6 billion people) are outside of the United States, Canada, China, and Western Europe. If the direction of AI is likely to have major consequences on their livelihoods and freedoms, it is worrisome that these many people have no voices in the trajectory of global AI technology. Establishing sufficient guardrails for AI rollout in their countries is thus especially important to avoid importing AI as an inappropriate technology. It is thus crucial to ensure that developing nations' voices are heard in the direction of global AI research.

It is unclear who will speak for these 6 billion people in the developing world. In an ideal world, governments or a consortia of governments could be candidates, but a large fraction of this population lives under authoritarian governments where leaders do not often defend regular people's interests.

8.6 Concluding Remarks

Including robust worker voices and developing country perspectives in the debates shaping the future of AI is not just a matter of fairness. The true promise of AI is its potential to empower workers and citizens, even if this promise is far from being realised. Such a promise necessitates detailed information about the needs, competencies, and perspectives of the people who will actually use the technologies. Yet whether key decision makers decide to listen to those who will be affected—and exactly whose voices are heard—is a political choice. At present, both the current ideology of the AI sector that sees data, computational capacity, and the genius of tech leaders as the most important inputs for the advancement of AI, as well as the financial interests of leading companies and their executives, do not favour broadening the set of voices and power centres influencing the future of this promising technology.

Broadening the set of stakeholders heard, although imperative, will not be easy. The challenge ahead is made more difficult by the fact that neither workers in the industrialised world nor workers and citizens around the developing world are particularly well organised. The labour movement has been in decline in much of the West for at least 5 decades, and this decline has accelerated in many countries, with a few attempts to unionise new sectors notwithstanding.²⁸ The developing world has rarely spoken with a common voice on any topic, especially one that runs counter to the interests of the United States and China. China's success in co-opting the BRICS grouping by expanding it to include its allies does not bode well for the hope that independent developing nations will be able to come together and forge a unified message on how AI can be best used and developed to help their citizens.²⁹

There is a way forward, however. Attempts to change the current equilibrium in the field of AI and the current dominance of a few tech companies and executives

are likely to be synergistic with other changes. For example, if democratic control over some of the AI functions occurs in the United States (e.g. if lawmakers pass legislation that limits the ability of tech companies to collect and to monetise data), then this can also pave the way for international regulation as tech companies would be weaker to resist reasonable demands from the developing world. Likewise, if worker voices become more robust, this could also facilitate the formation of a block of developing nations to articulate a common perspective on AI.

It would also be useful to develop several common perspectives that can make the voices of both workers and developing nations more effective. Centring the discussion on having a pro-worker and pro-citizen direction for AI or an anti-worker, anti-democracy one is a useful starting point. Once different parties in the conversation—including tech companies—accept that there is such a choice and that the pro-worker, pro-democracy direction is both technically feasible and socially desirable, it becomes much harder to pursue the current direction of research that has already caused harm for workers, equality, democracy, and well-being.

These debates could also broaden the scope of the discussion. For example, absent in current discussions is the question of whether foundation models (i.e. where resources are being spent to develop human-sounding communication capabilities in general interactions) are the best architecture for pro-worker AI tools that provide useful information to human decision makers. Any success by workers from industrialised nations in pushing tech companies to develop more tools that can be useful to human workers, and any brakes on excessive automation, would also help the developing world.

AI governance is about politics, and if the power to shape the direction of AI remains askew, the fruits of this new technological platform are unlikely to benefit most people. The antidote is multipolar AI governance that aims to produce a more inclusive political process. This will allow a richer set of voices to shape the future of AI in a way that is consistent with a broader set of objectives and interests.

Notes

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1 For example, an evaluation of the medium-term microeconomic effects of recent AI advances and examination of many forecasts that exaggerate the potential of AI are provided in Acemoglu (2024).

2 One famous contribution to the discourse from AI experts, which led to the departure of two notable Google researchers, is Bender et al. (2021). For an overview of more recent developments, see Capraro et al. (2024).

3 For example, see Huttenlocher, Ozdaglar, Goldston (2023) and Alam et al. (2024).

4 See as follows for more discussion of the Communist Party's control over direction of AI, as well as Chapter 10 on Chinese digital surveillance in Acemoglu and Johnson (2023b).

5 See Acemoglu and Johnson (2023a) and Acemoglu, Autor, Johnson (2023).

6 See Acemoglu and Autor (2011), Autor and Dorn (2013), and Autor (2019).

- 7 About 50%–70% of changes in the US wage structure since 1980 can be explained by the decline of work involving routine tasks in rapidly automating industries (Acemoglu and Restrepo, 2020, 2022).
- 8 See Acemoglu and Restrepo (2018, 2019).
- 9 Current AI–robotics integration research challenges include improving robot control, perception, decision-making, path planning, simultaneous localisation and mapping (SLAM), reinforcement learning, sensor fusion methods, and other topics. For an overview of large language models for robotics, see Zeng et al. (2023). For more on autonomous navigation, see Nahavandi et al. (2022). For additional AI-driven research questions related to multi-goal reinforcement learning, see Plappert et al. (2018).
- 10 In Acemoglu and Johnson (2023a), Jürgen Habermas’s notion that ‘the public sphere, defined as forums where individuals form new associations and discuss social issues and policy, is pivotal for democratic politics . . . [by allowing] people to freely participate in debates on issues of general interest without a strict hierarchy based on preexisting status’ was noted (p. 372). In the early 2000s, social media held the promise of creating a new public sphere, but filter bubbles, content moderation failures, and other missteps largely squandered the opportunity. See also Habermas (1991).
- 11 See Acemoglu and Johnson (2023a: Chapter 10) and Weyl and Tang (2024).
- 12 For more discussion of the damages to democracy and mental health, see Acemoglu (2023a).
- 13 See Acemoglu and Johnson (2023a: Chapter 9); Acemoglu (2021); and Acemoglu and Restrepo (2019).
- 14 See Acemoglu and Johnson (2023a: Chapter 9) and Acemoglu, Jordan, Weyl (2021).
- 15 For more discussion on AI’s tendency towards oligopoly, see Acemoglu and Johnson (2023b).
- 16 For more on the US perspective, see Zuboff (2019). For the China perspective, see Lei (2023).
- 17 For example, consider the challenges that Jack Ma has faced with Alibaba and the Ant Group after criticising China’s regulatory system, or the scrutiny of DiDi Global’s ride-hailing platform under China’s national security and cybersecurity laws (Mulrenan, 2021). For more on China’s comparative advantage in surveillance AI technology driven by governmental interests, see Beraja et al. (2023).
- 18 For more ideas on accomplishing changes in norms, see Acemoglu and Robinson (2023).
- 19 Amongst the more notable pieces of legislation protecting consumer data, privacy, and competition, the European Union paved the way with the General Data Protection Regulation in 2018 and Digital Markets Act and Digital Services Act in 2022.
- 20 This has not been the standard practice of labour unions in the United States; it is far more common to negotiate on the terms of work schedules, compensation, and benefits.
- 21 For more on the accomplishments of organised labour in the automotive industry of the 1950s and 1960s, see Noble (1984). For more discussion on the implications of the 2023 Writers Guild of America (WGA) strike, see Acemoglu, Johnson, Lentsch (2023). See also an overview of the AI provisions in the ratified WGA–Alliance of Motion Picture and Television Producers agreement (WGA West, 2024).
- 22 See Acemoglu and Johnson (2023a: Chapter 7) for further discussion of the US and Swedish cases of organised labour and their sway over the broader issues of work arrangements. Germany’s case is outlined in Jäger, Noy, Schoefer (2022).
- 23 It is not an uncommon—nor recent—problem for the international aid community. See, for example, Gibson et al. (2005).
- 24 See Beraja et al. (2023). For an interactive map showing the global partnerships and relationships between Chinese AI and surveillance companies and various governments, companies, and universities, see ASPI. “Mapping China’s Tech Giants.” <https://chinatechmap.aspi.org.au/#/map/>

- 25 While Chinese companies supply surveillance technology to 63 countries, US companies also provide AI surveillance tech to 32 countries, with IBM (11 countries), Palantir (9 countries), and Cisco (6 countries) leading the way (Feldstein, 2019). The AI Global Surveillance (AIGS) index used in Feldstein (2019) suggested that the United States uses smart/safe city tech, facial recognition, and smart policing tools, including some from China. For more discussion on the use of surveillance and monitoring tools by the United States and the European Union, see Sahin (2020).
- 26 For example, the snooping technology Pegasus, developed in Israel and made available as a ‘zero-click’ Trojan horse virus, has been used extensively in India and Mexico. See Acemoglu and Johnson (2023a: Chapter 10) for more discussion. On the use of Pegasus in India, see Singh (2021); Priest, Timberg, Mekhennet (2021); and Bergman and Mazzetti (2021).
- 27 For more on the rise of worker tracking in firms like Amazon, JP Morgan Chase, McDonald’s, and UnitedHealth, see Kantor et al. (2022).
- 28 For more discussion on the current state of unions and the collective action problem that precludes greater unionisation rates—even while pro-union sentiment is relatively high—see Naidu (2022). For further analysis of de-unionisation, see Farber et al. (2021).
- 29 See Acemoglu (2023b).

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